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# Hawkeye Replacement Policy in ChampSim

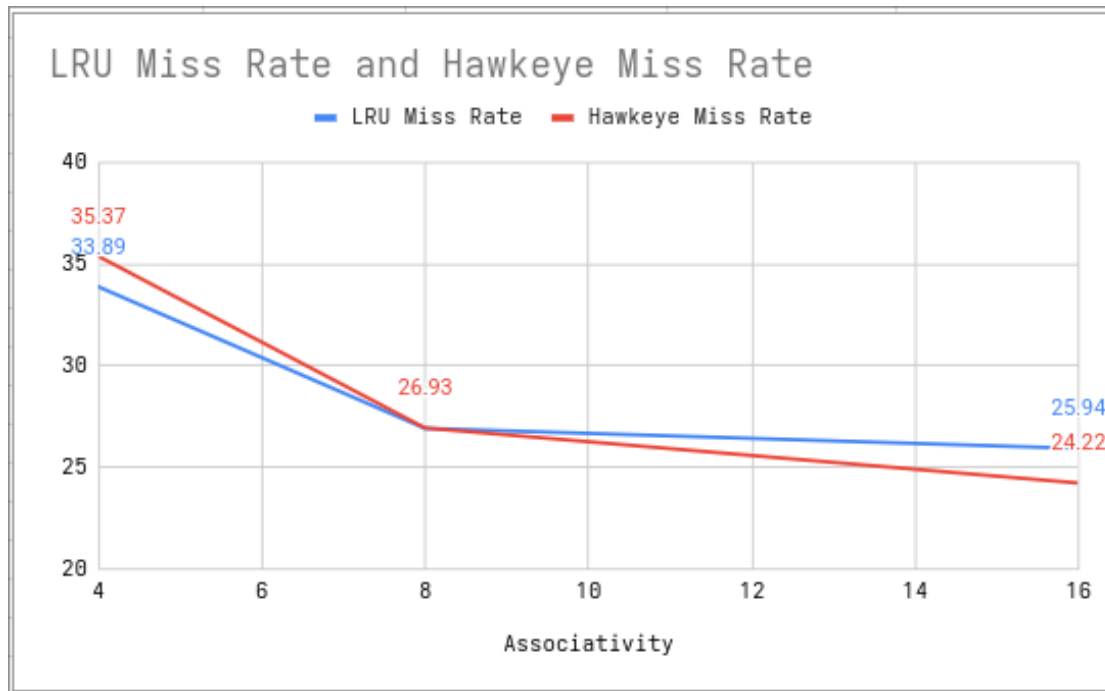
E0 - 243, HPCA

## Learnings/Observations:

1. Hawkeye's advantage over LRU on 456.hmmer is not uniform across associativity

| Config        | Policy        | ACCESS            | MISS  | MISS_RATE |
|---------------|---------------|-------------------|-------|-----------|
| 4-way         | LRU           | 265856            | 90091 | 33.89     |
| 4-way         | Hawkeye       | 265859            | 94022 | 35.37     |
| 8-way         | LRU           | 265798            | 71579 | 26.93     |
| 8-way         | Hawkeye       | 265798            | 71568 | 26.93     |
| 16-way        | LRU           | 265810            | 68963 | 25.94     |
| 16-way        | Hawkeye       | 265857            | 64391 | 24.22     |
|               |               |                   |       |           |
|               |               |                   |       |           |
| Associativity | LRU Miss Rate | Hawkeye Miss Rate |       |           |
| 4             | 33.89         | 35.37             |       |           |
| 8             | 26.93         | 26.93             |       |           |
| 16            | 25.94         | 24.22             |       |           |

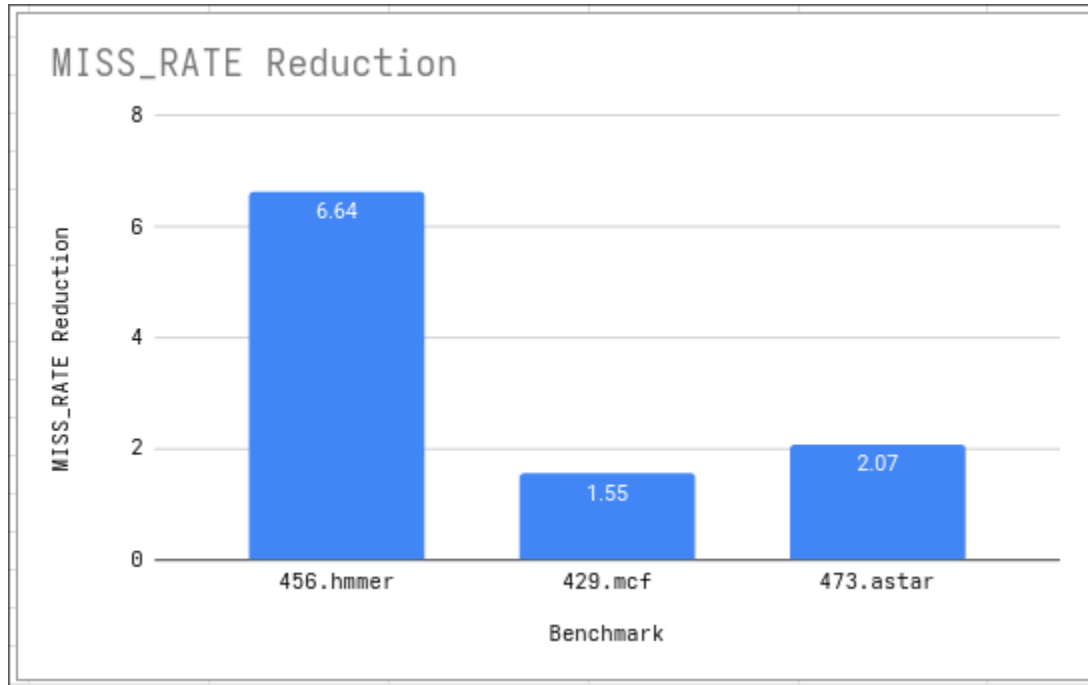
As observed, the benefits of Hawkeye grow with associativity, which is consistent with the findings from (Jain & Lin, 2016). Where this implementation strays from the paper is in its simplistic implementation, which results in Hawkeye dipping below LRU at the 4-way level.



## 2. OPTgen's usage-interval logic reproduces Belady's optimal decisions exactly on the paper's own worked example

(Jain & Lin, 2016) confirmed the accuracy of the standalone OPTgen implementation before relying on the full system results. Here, the software is compared against the specific example provided in Section 3.1 & Figure 6 of the source document. The test utilized the access sequence A, B, B, C, D, E, A, F, D, E, F, C and a simulated cache capacity of 2 lines. In (Jain & Lin, 2016) the standard data indicates that 4 hits occur - those hits happen during the second accesses to B, A, D & F, while the second accesses to E & C result in misses. The implemented OPTgen code generated the same outcome. It assigned the matching HIT/MISS label to every individual access and reached the same total of 4 hits. By achieving this result, we ensured that the occupancy vector mechanism functioned according to the described algorithm - this mechanism includes usage interval tracking, capacity threshold comparison and miss-bypass-without-update. It was necessary to verify those components before the authors used the mechanism to operate the predictor or RRIP components.

3. 429.mcf did not perform as per the paper's results, it was in fact the weakest benchmark in this implementation



In (Jain & Lin, 2016), 429.mcf is highlighted as one of Hawkeye's best-performing benchmarks. In this implementation, it showed the smallest miss-rate reduction of the 3 benchmarks, despite having the highest baseline LLC miss rate(68.42% for LRU vs 25.94% and 19.63%). A possible explanation may be that .mcf misses are dominated by genuinely irregular, low-locality accesses that no history-based predictor can convert into hits, whereas the much lower miss rates of the other 2 may leave some learning avenue for predictions to exploit.

| Benchmark | LRU ACCESS | LRU MISS | LRU MISS_RATE | HAWKEYE ACCESS | HAWKEYE MISS | HAWKEYE MISS_RATE | MISS_RATE Reduction |
|-----------|------------|----------|---------------|----------------|--------------|-------------------|---------------------|
| 456.hmmer | 265810     | 68963    | 25.94         | 265857         | 64391        | 24.22             | 6.64                |
| 429.mcf   | 1943704    | 1259882  | 64.82         | 1943656        | 1240160      | 63.81             | 1.55                |
| 473.astar | 658152     | 129217   | 19.63         | 657582         | 126453       | 19.23             | 2.07                |

